

Particle Physics — Interactions and Fields

IMAPP lecture notes

Spring 2026

1 Classical potential picture

For a static source, the potential satisfies a Klein-Gordon-type equation,

$$\left(\nabla^2 - \frac{1}{R_0^2}\right) V(\mathbf{R}) = \delta^{(3)}(\mathbf{R}).$$

For a massless mediator ($R_0 \rightarrow \infty$), this gives the Coulomb form $V(R) \propto 1/R$. For a massive mediator, the potential is Yukawa-like:

$$V(R) \propto \frac{e^{-R/R_0}}{4\pi R}, \quad R_0 \simeq \frac{1}{m_X}.$$

The finite range of the strong nuclear force, roughly $R_0 \sim 1$ fm, corresponds to a mediator mass of order 200 MeV.

2 Exchange of a virtual particle

At higher energies, interactions are described by quantum field theory. A scattering amplitude contains the couplings at the two vertices and the propagator of the exchanged particle:

$$\mathcal{M} \sim g_a \frac{1}{q^2 - m_X^2} g_b,$$

where q is the transferred four-momentum. The exchanged particle is virtual and need not satisfy $q^2 = m_X^2$; it is off shell.

3 Feynman channels

For $2 \rightarrow 2$ scattering, the momentum transfer can be organized into channels:

- t -channel exchange: $q = p_1 - p_3$, typically spacelike ($q^2 < 0$).
- s -channel exchange: the incoming particles form a time-like intermediate state ($q^2 = s > 0$).
- u -channel exchange: the crossed version of the process.

4 Interaction strengths

The electromagnetic interaction is controlled by the fine-structure constant $\alpha \simeq 1/137$. Strong interactions are characterized by colour charge and a scale-dependent coupling. Weak interactions involve flavour and/or neutrino changes and are suppressed at low energy by massive-boson propagation, with a typical factor proportional to $1/M_W^2$.

Summary

The potential picture becomes a field-theoretic exchange picture: range is set by mediator mass, amplitudes contain propagators, and scattering is classified by s , t , and u channels.